

DEPARTMENT OF ECE**BASIC PYTHON PROGRAMMING**

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Data Manipulation in Data Sets using Python**Abstract**

Data manipulation is a critical step in the data analysis process that involves transforming raw data into a meaningful and structured format. With the rapid growth of data in various domains, efficient data handling techniques have become essential. This document focuses on data manipulation using Python, highlighting key operations such as data cleaning, filtering, sorting, and aggregation. Python, along with its powerful libraries like Pandas, provides a flexible and efficient environment for handling large datasets. The study also demonstrates simple programs to perform common data manipulation tasks and discusses real-time applications in fields like education, healthcare, agriculture, and business analytics. Effective data manipulation improves data quality, enhances decision-making, and supports advanced analytics and machine learning systems.

Introduction

Data manipulation is the process of converting raw data into a structured and useful format. In real-world scenarios, data is often incomplete or unorganized, making it difficult to analyze directly. Python is widely used for data manipulation because of its simplicity and powerful libraries like Pandas, which allow efficient handling of large datasets.

Explanation of Data Manipulation

Data manipulation includes basic operations such as:

- Cleaning missing or incorrect data
- Filtering required information
- Sorting data for better understanding
- Performing calculations like average or sum

These operations help in preparing data for analysis and decision-making.

Program 1: Basic Data Manipulation

```
import pandas as pd
```

```
# Creating dataset
```

```
data = {  
    "Name": ["A", "B", "C", "D"],  
    "Marks": [85, 90, None, 70],  
    "City": ["Hyderabad", "Delhi", "Hyderabad", "Mumbai"]  
}
```

```
df = pd.DataFrame(data)
```

```
# Handling missing values
```

```
df["Marks"].fillna(df["Marks"].mean(), inplace=True)
```

```
# Filtering students with marks > 80
```

```
top_students = df[df["Marks"] > 80]
```

```
# Sorting data
```

```
sorted_df = df.sort_values(by="Marks", ascending=False)
```

```
print("Updated Data:\n", df)
```

```
print("\nTop Students:\n", top_students)
```

```
print("\nSorted Data:\n", sorted_df)
```

Program 2: Real-Time Dataset Example

```
import pandas as pd
```

```
# Reading dataset
```

```
df = pd.read_csv("students.csv")
```

```
# Removing missing values
```

```
df.dropna(inplace=True)
```

```
# Calculating average marks
```

```
average_marks = df["Marks"].mean()
```

```
# Finding students who failed
```

```
failed_students = df[df["Marks"] < 40]
```

```
print("Average Marks:", average_marks)
```

```
print("\nFailed Students:\n", failed_students)
```

Real-Time Applications

- **Education:** Student performance analysis and grading
- **Healthcare:** Patient data cleaning and analysis
- **Agriculture:** Crop and weather data analysis
- **Business:** Sales and customer data processing

- **Machine Learning:** Preparing clean datasets for models

Conclusion

Data manipulation is an essential step in working with data. Python makes this process simple and efficient through powerful libraries like Pandas. By cleaning, organizing, and transforming data, users can extract meaningful insights and support better decision-making. These techniques are widely used across industries and form the foundation for data analysis and machine learning applications.